Easy-to-Hard Structure for Remote Sensing Scene Classification in Multitarget Domain Adaptation

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***Abstract*— Multitarget domain adaptation (MTDA) is a transfer learning task that uses knowledge extracted from a labeled source domain to adapt across multiple unlabeled target domains. The MTDA setting is more complicated than the single-source-single-target domain adaptation (S^{3}TDA) setting because domain shift not only exists in each pair of a source–target domain but also exists among different target domains. In addition, multiple-target domains have their own unique characteristics because they are often collected from various conditions. The semantic information in each target domain can be damaged when they are naïvely merged into a single-target domain. Therefore, the trained model struggles to distinguish between representations in the combined target domain, which degrades the classification performance. Furthermore, the knowledge transferability from the source domain to multiple-target domains in prior studies leaves room for improvement because they only focus on exploiting the relationship of source–target pairs while failing to consider the correlation among multiple-target domains. This article introduces an easy-to-hard adaption structure to solve these problems in MTDA. The proposed method consists of three components: Extracting source representations, Hierarchical intratarget feature Alignment, and Collaborative intertarget feature Alignment, called EHACA. These components are used to encode the semantic information in each target domain and explore the relationships between the source and target domains, and among different target domains. The proposed method shows outstanding classification performance over five remote sensing datasets of MTDA tasks, surpassing state-of-the-art approaches in most experimental scenarios.**

***Index Terms*— Collaborative learning, deep learning, domain adaptation (DA), easy-to-hard, multiple-target domains, remote sensing (RS) image classification.**

NOMENCLATURE

$\mathcal{G}$	Shared feature extractor.
$\mathcal{F}_j$	jth classifier.
$\mathcal{D}_S$	Dataset of the source domain.
$\mathcal{D}_{T_j}$	Dataset of the jth target domain.
x_S^i	ith labeled sample in the source domain.
y_S^i	Label of the ith sample in the source domain.
n_S	Number of samples in the source domain.
$x_{T_j}^i$	ith unlabeled sample in the target domain j.
n_{T_j}	Number of samples in the target domain j.
$x^{i,w}$	Weakly augmented version of the ith sample.
$x^{i,\text{str}}$	Strongly augmented version of the ith sample.

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I. INTRODUCTION

TRANSFER learning is a cornerstone of deep learning, in which the knowledge of a model is utilized to train another model. To be specific, the weights of a model pretrained on a massive dataset are reused as the initial weights of a new model. Most transfer learning approaches assume that the training and test sets are collected from the same domain and, hence, share the distribution. However, in real-world applications, the training data distribution (source domain) usually differs from that of the test data (target domain). Thus, the classification performance on the target domain has room for improvement when we directly adopt the learned model on the source domain to the target domain due to domain shift [1] or unseen categories in the target data [2], [3]. To tackle this problem, domain adaptation (DA) [4], [5], [6], [7], [8], [9], [10], [11] emerges as the best solution that leverages both the labeled source samples and unlabeled target samples to minimize the domain shift by aligning source and target representations into the joint latent space. Recently, several methods have successfully used deep DA for remote sensing (RS) to address many applications, such as classification [4], [5], semantic segmentation [6], [7], ship detection [8], [9], and building damage detection [10], [11]. In these methods, the knowledge in a given source domain is extracted to transfer to a single-target domain called the "single-source-single-target" (S^{3}T) setting. However, S^{3}T is less attractive in real-world scenarios. As the number of target domains increases (new target domains are available), the model needs to be retrained for adaptation. Indeed, RS images have significantly different distributions because they are collected under various conditions, such as geographic locations, reflectance, illumination conditions, and the differences in satellites' sensor settings. Thus, the concept of a single-source domain simultaneously adapting to multiple-target domains—called multitarget DA (MTDA)—has recently received more attention for practical scenarios. Compared to the S^{3}T setting, the MTDA setting is much more challenging because domain shift not only exists in

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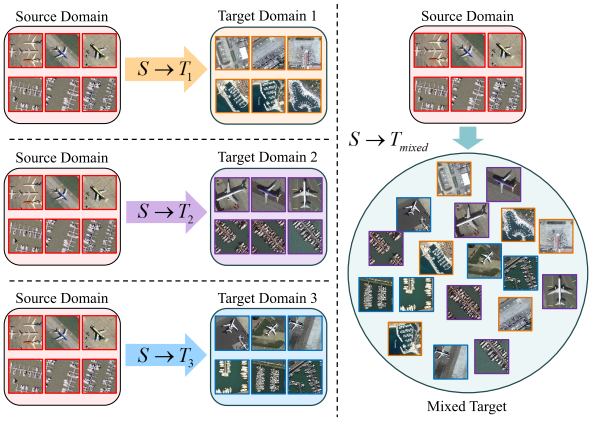


Fig. 1. Left-hand side column describes the operation of multiple *single-target* DA tasks; we need to retrain a model independently following a new target domain. The right-side column illustrates the operation in the *target-combined* setting.

each source–target pair but also exists among different target domains.

Prior S^{3}T methods [12], [13], [14], [15] can be extended to apply to the MTDA setting following two directions. The first direction is when a model, consisting of a feature extractor and a single classifier, is trained on each source–target pair independently, called multiple single-target DA (MSTDA), as shown in the left-hand side column of Fig. 1. However, this solution is not feasible in practical applications because the model needs to be retrained whenever a new target domain appears. Moreover, in this way, a trained model cannot simultaneously cover all representations across multiple-target domains, and the correlation among multiple-target domains is ignored. To solve these problems, in the second direction [16], [17], [18], they use data grouping-based methods in which all target domains are naïvely combined into a single-target domain, and then, the model can be trained on the source and combined target samples by utilizing exiting S^{3}T approaches, so the learned model can simultaneously obtain representations from the source and multiple-target domains.

However, a recent study [18] showed that multiple-target domains are collected from various conditions consisting of unique data characteristics and typically having different data distributions. Therefore, naïvely combining all multitarget domains as a single target can negatively impact the DA and classification performances on the target domain due to negative transfer. Furthermore, the trained model can struggle to encode all multitarget representations in the combined target data because it increases the data variance. To capture all features from this combined target data, we need to increase the size of the training model by adding more convolutional layers or increasing the feature dimension in each layer, which leads to an increase in the model complexity and training time. Sharing the viewpoint of [18], [19] argues that the semantic information in different datasets can be damaged if we naïvely combine them. In addition, most existing MTDA approaches only focus on exploiting the relationship between source and target domains without considering exploring the correlations among different target domains.

The "divide-and-conquer" strategy [20] demonstrates that the variance in the training data can be reduced by separating it into multiple subsets in the division stage. Then, they trained the model to construct the entire embedding space by joining the specific embedding spaces of different data subsets together in the conquer stage. In addition, the advantage of the "divide-and-conquer" strategy is also well verified in a recent study RSSC-ETDL [21] with the multiview manner. To be specific, RSSC-ETDL first splits the original dataset into multiple nonoverlapped subsets. Then, the semantic information of these subsets is encoded by the different convolutional neural network (CNN) models sharing the same architectural style. These CNN models work as multiple viewers to capture a partial feature of the original dataset. Finally, the partial information of multiple views is adaptively combined. By doing so, the trained model not only effectively exploits training data for generalization but also alleviates the negative impact of incorrect labels. Therefore, the classification performance is significantly improved. Inspired by these aforementioned approaches, we proposed an effective and flexible method to solve challenges in the MTDA setting to preserve unique characteristics in each target domain and explore the correlation among different target domains. The proposed method equips three training components: ***E**xtracting source representations*, ***H**ierarchical intratarget feature **A**lignment*, and ***C**ollaborative intertarget feature **A**lignment*, called EHACA, which are conducted from easy to hard in terms of extracting the information manners of the training data to successfully enhance the generalization on the cross-target domain. Fig. 2 expresses the details of this strategy. Corresponding to the training procedure, we designed a novel framework consisting of a shared feature extractor $\mathcal{G}$ and multiple classifiers $\{\mathcal{F}_j\}_{j=1}^{N}$, where N is the number of target domains. Fig. 3 illustrates the architecture of the proposed EHACA.

As shown in Fig. 2, the *extracting source representations* is the easiest process indicated by the green color, where $\mathcal{G}$ and $\{\mathcal{F}_j\}_{j=1}^{N}$ can explicitly encode the representations of source samples based on label information via the standard cross-entropy loss. Then, the knowledge acquired from the source domain can be used for the next downstream tasks, such as *hierarchical intratarget feature alignment* and *collaborative intertarget feature alignment*.

The *hierarchical intratarget feature alignment*, as indicated by the orange color in Fig. 2, includes minimizing class confusion and contrastive learning strategies that are used to support different classifiers for obtaining the private information in each target domain. Specifically, $\mathcal{F}_j$ is first trained to highlight the high confidence score using the minimizing class confusion strategy. Then, we utilize the contrastive learning strategy for exploiting the information from the more complicated target samples to provide certainty of the prediction. By doing this, the different classifiers are trained on the different target domains to work as domain-expert classifiers.

Finally, the *collaborative intertarget feature alignment* is the most challenging task that is used to explore knowledge from other target domains for cross-target domain generalization. We propose a *collaborative learning* strategy for intertarget

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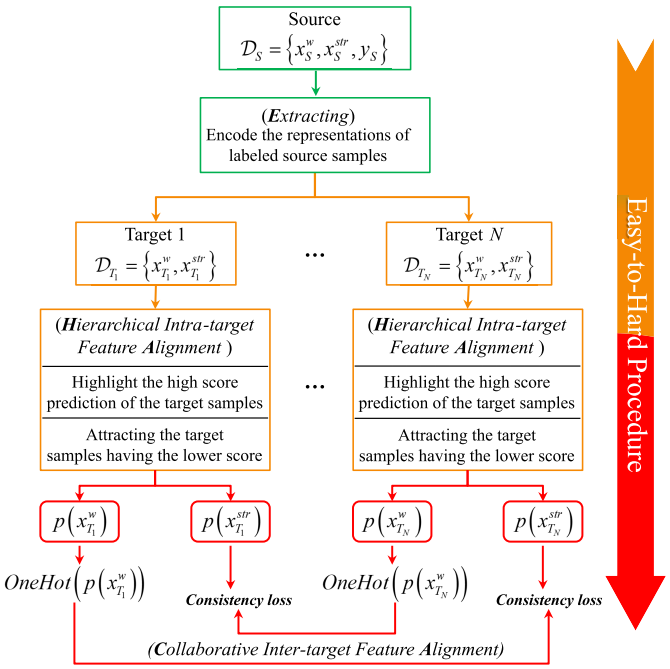


Fig. 2. Proposed easy-to-hard structure (EHACA).

feature alignment, which is indicated by red in Fig. 2. This training strategy allows different classifiers to exchange their knowledge to obtain a joint model for all target domains. To be specific, a domain-expert classifier receives complementary information of $(N-1)$ remaining target domains via the proposed collaborative process.

We summarize this article's main contributions as follows.

1) We introduce a novel method to solve existing challenges for MTDA in RS classification tasks. The proposed method successfully adapts the extracted knowledge of a single-source domain to multiple-target domains without requiring any additional computation of adversarial objectives, such as domain discriminators. Therefore, the proposed method's training procedure is simple and stable for real-world RS scene classification.
2) We propose a novel method for the MTDA task that aims to simultaneously find a joint latent space for source and multiple-target domains, and preserve private statistics in each target domain. Therefore, the model trained by our method can simultaneously alleviate the negative transfer effect across different domains and increase the discriminability of representations within each target domain.
3) To the best of our knowledge, the proposed EHACA is the first approach that takes advantage of the collaborative learning scheme to solve problems in the MTDA task of RS classification. Extensive experimental results show that the proposed method can generalize to the multitarget domain and significantly outperform state-of-the-art (SOTA) methods.

II. Related Work

Most previous DA methods focused on the S^{3}T DA (S^{3}TDA) [12], [13], [14], [15]. In contrast, a few works have examined MTDA [16], [17], [18], [22], [23], [24], [25] because it is more challenging; this section overviews research topics related to the S^{3}TDA and MTDA settings. In addition, we clarify the relationships between the proposed EHACA and previous approaches.

A. Single-Source-Single-Target Domain Adaptation

The goal of S^{3}TDA approaches [12], [13], [14], [15] is to adapt the discriminative model trained on a source domain that contains rich label information to a single-target domain without target label information. The adaptation strategy for these methods aims to minimize distributions between source and target domains; S^{3}T approaches can be categorized into two main groups: maximum mean discrepancy (MMD)-based methods [12], [13] and adversarial training-based methods [14], [15]. The first group aims to minimize the marginal distribution between source and target domains measured using the MMD metric. This metric estimates the similarity between two distributions based on the mean deviations of each pair class from the two different domains. The second group of S^{3}T methods uses the adversarial learning strategy to align source and target distributions. In these methods, the feature extractor is trained to simultaneously maximize domain confusion loss while minimizing classification loss to achieve domain invariance and class discrimination. However, these S^{3}TDA approaches only consider exploiting the relationship between a source–target domain pair. Therefore, extending these approaches to the MTDA task often fails because it is a suboptimal problem.

B. Multiple-Target Domain Adaptation

Several methods are proposed to address the challenges in the MTDA task, which can be categorized into two subgroups: *target-combined* DA [16], [17], [18] and *multitarget* DA settings [22], [23], [24], [25]. In *target-combined* DA, the knowledge from a single-source domain is used to transfer to mixed multiple-target domains. However, each target domain is collected under different conditions, so they contain various unique characteristics. Therefore, naïvely merging them into one can damage the semantic information in each target domain, leading to the trained model struggling to classify representations in these domains. In contrast, in *multitarget* DA scenarios, the DA model focuses on exploiting the relationship in each source–target pair to capture domain-specific features in each target domain. For example, Nguyen-Meidine et al. [23] use multiple teacher models to obtain representations for each target domain. However, this method ignores the correlation across different target domains. Moreover, using multiple CNNs to capture the private target information in each target domain leads to a huge increase in the model complexity.

C. Easy-to-Hard Strategy

In this section, we clarify the relationship between the proposed EHACA, Domain-aware Curriculum Graph Co-Teaching (D-CGCT) [22], collaborative consistency learning (CCL) [24], and multiple-view collaborative learning

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(MVCL) [26]. These approaches share a similar philosophy but contain different roles.

1) Relation to D-CGCT [22]: The level of domain shift of different target domains is not the same as the source domain. D-CGCT borrowed the easy-to-hard transfer strategy (EHTS) of [27] to develop the concept of easy-to-hard domain selection (EHDS). This approach focuses on the "different levels of domain" that gradually adapts from the easiest target domain to the hardest one by using multiple stages with a single model. At each stage, this model only considers feature alignment in a source–target pair. D-CGCT did not explore the correlation among different target domains. Therefore, it could have a suboptimal issue in which the trained model was more focused on a closer target. In contrast, we proposed an easy-to-hard structure focused on the "different levels of the sample." Specifically, the proposed model extracts easily knowledge of the source domain thanks to label information. Then, *hierarchical intratarget feature alignment* and *collaborative inter-target feature alignment* are conducted sequentially depending on the difficulty of the prediction on target samples to achieve cross-domain representation alignment at the category level.

2) Relation to CCL [24]: Similar to the proposed method, CCL uses the collaborative learning strategy for MTDA. However, CCL focuses on addressing the semantic segmentation task, while the proposed method uses the collaborative learning strategy for the RS classification task. Moreover, this approach utilizes Kullback–Leibler (KL) divergence to exchange knowledge among different domain-specific experts. In contrast, the proposed EHACA uses pseudolabeling and consistency regulation techniques.

3) Relation to MVCL [26]: The main concept of MVCL focuses on exploiting unlabeled information by exploring the correlation between two views from two different models, called cotraining. In contrast, collaborative learning in the proposed EHACA leverages information learned from a group—including multiple expert models—to benefit the training of the specific model. The cotraining strategy can be considered a case study of collaborative learning when there are two source domains. However, our goal is to develop a novel algorithm for the MTDA task with an unlimited number of target domains.

III. Proposed Method

In this section, we first provide the notations and problem definitions of the MTDA task. Then, we briefly express our solution to solve the problems of the MTDA task in detail.

A. Problem Definition

In the MTDA setting, we have a given labeled training dataset of a source domain, $\mathcal{D}_S = \{(x_S^i, y_S^i)\}_{i=1}^{n_S}$ that includes pairs of the source image x_S^i associated with the label y_S^i, and $\{T_j\}_{j=1}^N$ unlabeled target domain datasets, where N is the number of target domains. The target domain j has $\mathcal{D}_{T_j} = \{(x_{T_j}^i)\}_{i=1}^{n_{T_j}}$ unlabeled samples, where $x_{T_j}^i$ is the ith sample of an unlabeled image in the target domain $j \in N$. n_S and n_{T_j} are the numbers of image samples in the source domain and the target domain j, respectively. The goal of MTDA is to predict the labels of multiple-target domains by exploiting the information of a single labeled source domain and unlabeled target domains under two basic assumptions: 1) samples from the source and target domains share the same label space $y_S = \{y_{T_j}\}_{j=1}^N$, $y_S \in \mathbb{R}^K$, where K is denoted as the number of classes, and $\{y_{T_j}\}_{j=1}^N$ are only used in the testing phase for evaluation and 2) the distributions of the multiple-target domains are different, and the distribution in each target domain is different from the source distribution. Nomenclature summarizes the notations and definitions frequently employed in this article.

To enhance the model's generalizability, we conducted stochastic data augmentation, which includes a set of transformation techniques, to extend the given training set. Specifically, we used the weak and strong augmentation strategies to create two weak and strong augmentation versions for a given input $x \in \{\mathcal{D}_S, \mathcal{D}_{T_1}, \mathcal{D}_{T_2}, \ldots, \mathcal{D}_{T_N}\}$. To generate the weakly augmented image, x^w, we fed the original image into the weakly augmented function, WeakAug(.), which includes simple transformation functions, such as random flipping and random cropping. To generate the strongly augmented image, x^{str}, we used StrongAug(.) that includes RandAugment [28]. For example, the ith labeled source sample, x_S^i, has two versions $x_S^{i,w}$ and $x_S^{i,\text{str}}$. Similarly, the unlabeled target samples in the jth target domain, x_{T_j}, has two versions $x_{T_j}^w$ and $x_{T_j}^{\text{str}}$. The training procedure for the proposed method is processed in detail as follows.

B. Training Procedure of the Proposed EHACA

1) Step 1 (Extracting Information From a Labeled Source): In this step, the information of source samples is extracted easily while minimizing classification loss to fit predictions of both weakly and strongly augmented source images with the same ground-truth label. This cost function is defined as follows:

$$\mathcal{L}_{\mathcal{F}_j}^{\text{Extract}} = -\mathbb{E}_{\left(x_S^{i,w}, y_S^i\right)}\left[y_S^i \log\left(\mathcal{F}_j\left(\mathcal{G}\left(x_S^{i,w}\right)\right)\right)\right] - \mathbb{E}_{\left(x_S^{i,\text{str}}, y_S^i\right)}\left[y_S^i \log \mathcal{F}_j\left(\mathcal{G}\left(x_S^{i,\text{str}}\right)\right)\right] \quad (1)$$

where $\mathcal{L}_{\mathcal{F}_j}^{\text{Extract}}$ is the cross-entropy loss to train classifier $\mathcal{F}_j$ and y_S^i is the label of $x_S^{i,w}$ and $x_S^{i,\text{str}}$, respectively. The cost function used to train the feature extractor $\mathcal{G}$ is calculated as follows:

$$\mathcal{L}_{\mathcal{G}}^{\text{Extract}} = \frac{1}{N}\sum_{j=1}^{N} \mathcal{L}_{\mathcal{F}_j}^{\text{Extract}}. \quad (2)$$

The parameters of $\mathcal{G}$ and $\{\mathcal{F}_j\}_{j=1}^N$ are updated by minimizing the classification error term to preserve the discriminability of the representations as in (1) and (2).

2) Step 2 (Hierarchical Intratarget Feature Alignment): This step focuses on extracting unique characteristics in each target domain for intratarget feature alignment. First, the target samples having a high confidence prediction are highlighted by utilizing the minimum class confusion (MCC) method inspired by Jin et al. [29] to reduce the class confusion of the extracted target representations by the pretrained model on the labeled source data. However, the original MCC loss function is used

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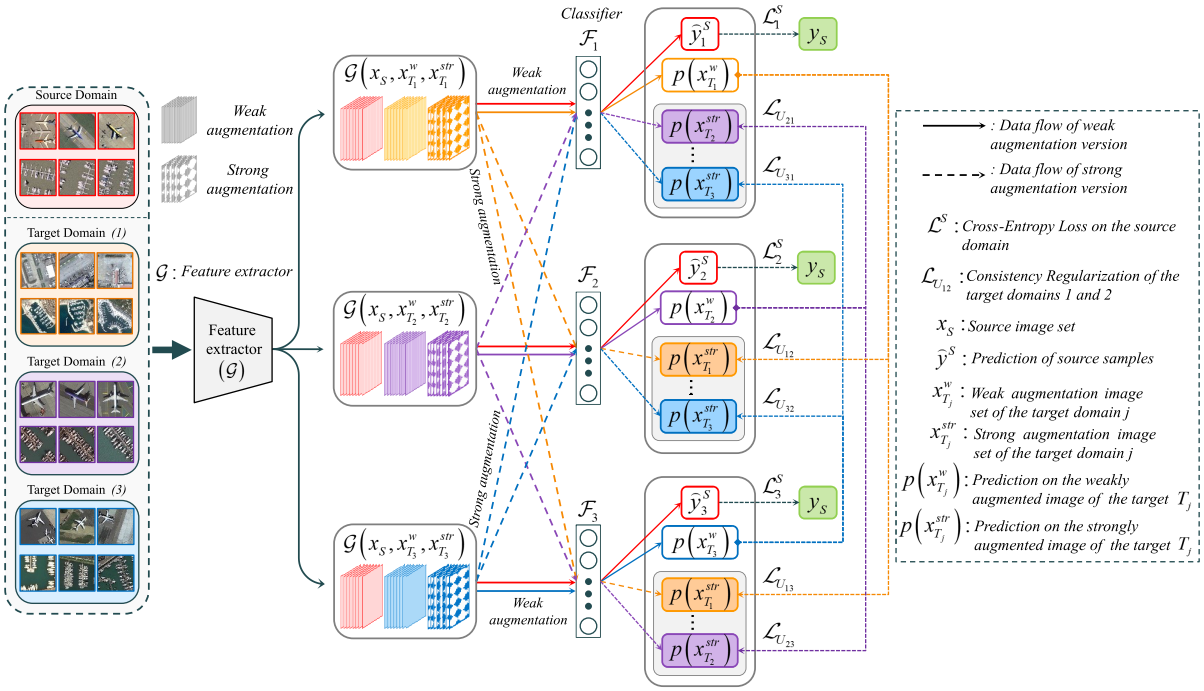


Fig. 3. Architecture of the proposed method for RS scene classification in MTDA with $N = 3$ (where N is the number of target domains).

for a single classifier on a single type of target data. Thus, we modified this loss to adapt to multiple classifiers and with various augmented target data. Specifically, we calculated the MCC losses on weak and strong augmentation images. As in [29], the *cross-class* correlation between the two predicted probabilities of classes k and k', where $k, k' \in K$, on the weakly augmented images in each batch is defined as follows:

$$C_{j,kk'}^{w} = \left(\hat{\mathbf{p}}_{j,k}^{w}\right)^{\top}\left(\hat{\mathbf{p}}_{j,k'}^{w}\right) \tag{3}$$

where $\hat{\mathbf{p}}_{j,k}^{w} \in \mathbb{R}^{B\times 1}$ and $\hat{\mathbf{p}}_{j,k'}^{w} \in \mathbb{R}^{B\times 1}$ are vectors of the soften probabilities for the kth and k'th classes of weakly augmented images in each batch, respectively, predicted by the classifier index j. B is the number of target samples in each batch. The class correlation in (3) evaluates the confusion level of the jth classifier to distinguish the kth and k'th classes in B samples.

To effectively classify ambiguous classes, we use the weighting strategy with the class correlation that highlights the samples having high confidence (large certainty) when categorizing target samples. To be specific, we utilized the entropy function $H(p) \triangleq -\sum p \log p$ to measure the uncertainty of the probability distribution p. Therefore, the entropy of the weak version of the ith target sample, as predicted by the jth classifier, is defined as follows:

$$H\left(\hat{p}_{j,k}^{i,w}\right) = -\sum_{k=1}^{K}\left(\hat{p}_{j,k}^{i,w}\right)\log\left(\hat{p}_{j,k}^{i,w}\right). \tag{4}$$

As expressed in (4), the samples with lower entropy values (larger certainty) in class predictions provide more reliability and contribute more to minimizing between-class confusion. Specifically, the important sample for reducing class confusion can be defined as follows:

$$W_j^{i,w} = \frac{B\left(1+\exp\left(-H\left(\hat{p}_{j,k}^{i,w}\right)\right)\right)}{\sum_{k'=1}^{K}\left(1+\exp\left(-H\left(\hat{p}_{j,k'}^{i,w}\right)\right)\right)} \tag{5}$$

where $W_j^{i,w}$ indicates the important weight of the ith weak sample for representing the class confusion by the jth classifier. $W_j^{i,w}$ receives a higher value, which implies that this sample is highlighted. Therefore, the class confusion in (3) can be calculated as follows:

$$\begin{aligned} C_{j,kk'}^{w} &= \left(\hat{\mathbf{p}}_{j,k}^{w}\right)^{\top}\mathbf{W}^{w}\left(\hat{\mathbf{p}}_{j,k'}^{w}\right) \\ &\left[= \hat{p}_{j,k}^{1,w} \quad \cdot \quad \hat{p}_{j,k}^{B,w}\right]\begin{bmatrix} W_j^{1,w} & 0 & 0 \\ 0 & \cdot & 0 \\ 0 & 0 & W_j^{B,w} \end{bmatrix}\begin{bmatrix} \hat{p}_{j,k'}^{1,w} \\ \cdot \\ \hat{p}_{j,k'}^{B,w} \end{bmatrix} \end{aligned} \tag{6}$$

where $\mathbf{W}^{w} \in \mathbb{R}^{B\times B}$ is the corresponding diagonal matrix whose diagonal elements are the weight factors of each sample in a batch. Finally, the cost function for minimizing class confusion on the weak augmentation version of the unlabeled target domain is computed as follows:

$$\mathcal{L}_{\mathcal{F}_j}^{w} = \frac{1}{K}\sum_{k=1}^{K}\sum_{k'\neq k}^{K} C_{j,kk'}^{w}. \tag{7}$$

By this, the representations between the different classes are more discriminative. Similarly, we minimized the cross-class confusion on the strong augmentation version by using the following loss:

$$\mathcal{L}_{\mathcal{F}_j}^{\text{str}} = \frac{1}{K}\sum_{k=1}^{K}\sum_{k'\neq k}^{K} C_{j,kk'}^{\text{str}}. \tag{8}$$

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Finally, the cost function to update the classifier $\mathcal{F}_j$ that minimizes class confusion is calculated as follows:

$$\mathcal{L}_{\mathcal{F}_j}^{\text{MCC}} = \mathcal{L}_{\mathcal{F}_j}^{w} + \mathcal{L}_{\mathcal{F}_j}^{\text{str}}. \tag{9}$$

The cost function to update the feature extractor $\mathcal{G}$ in this step is calculated as follows:

$$\mathcal{L}_{\mathcal{G}}^{\text{MCC}} = \frac{1}{N}\sum_{j=1}^{N} \mathcal{L}_{\mathcal{F}_j}^{\text{MCC}}. \tag{10}$$

However, to enhance the consistency of the trained model by covering more target information, we use contrastive learning [30], which encourages the model to provide a similar view of the different augmentations of the same image. Specifically, we consider the correlation between weak and strong versions of the same unlabeled target sample to improve the discriminative feature's ability. Without loss of generality, the loss function to optimize the pairwise similarity between the two augmented versions of a given image is defined as

$$\mathcal{L}_{\mathcal{F}_j}^{\text{Cont}} = -\log \frac{\exp\left(\text{sim}\left(\mathbf{z}_j^{i,w}, \mathbf{z}_j^{i,\text{str}}\right)/T\right)}{\sum_{\mathbf{z}^{i'} \in A(i)} \exp(\text{sim}(\mathbf{z}^i, \mathbf{z}^{i'})/T)} \tag{11}$$

where $\text{sim}(\cdot)$ is the cosine similarity function. T is denoted as a temperature parameter. $\mathbf{z}^i \in \{\mathbf{z}_j^{i,w}, \mathbf{z}_j^{i,\text{str}}\}$ is the similarity pair, in which $\mathbf{z}_j^{i,w}$ and $\mathbf{z}_j^{i,\text{str}}$ are output prediction vectors according to weakly and strongly augmented versions of a target image, x^i, predicted by the classifier, $\mathcal{F}_j$. $\mathbf{z}^{i'} \in A(i) = \{\mathbf{z}_j^{i',w}, \mathbf{z}_j^{i',\text{str}}\}_{i'=1,i'\neq i}^{n_{T_j}}$ is the augmentation set of dissimilarity samples. The output prediction vectors according to weakly and strongly augmented versions of the dissimilarity image, $x^{i'}$, predicted by the classifier $\mathcal{F}_j$, are $\mathbf{z}_j^{i',w}$ and $\mathbf{z}_j^{i',\text{str}}$, respectively. The model is trained to minimize $\mathcal{L}_{\mathcal{F}_j}^{\text{Cont}}$, which increases the similarity between two predictions $\mathbf{z}_j^{i,w}$ and $\mathbf{z}_j^{i,\text{str}}$, and dissimilarity of the rest predictions of augmentation samples. The cost function to update the feature extractor $\mathcal{G}$ in the contrasting step is calculated as follows:

$$\mathcal{L}_{\mathcal{G}}^{\text{Cont}} = \frac{1}{N}\sum_{j=1}^{N} \mathcal{L}_{\mathcal{F}_j}^{\text{Cont}}. \tag{12}$$

By minimizing the losses in (9)–(12), the trained model can extract the specific features in each target domain. Furthermore, the different classifiers obtain unique characteristics to provide a specific view of this target domain.

3) Step 3 (Collaborative Intertarget Feature Alignment): We identified that using only the intratarget feature alignment on an individual domain-expert classifier for self-training may fail to a suboptimal issue where the trained model can be unsatisfactory to become a joint model across all target domains, that is, because it lacks connections among different domain-expert classifiers to jointly explore information in various target domains. Therefore, we extend the cotraining [26] to collaborative learning. In [26], the two classifiers cooperate to teach each other using pseudolabeling. However, the core idea of our collaborative learning strategy is that multiple domain-expert classifiers that have different views collaborate to supervise the learning of a particular expert classifier for cross-view relation transfer. To achieve this goal, first, the mth classifier that is the domain-specific expert on the target domain m provides its predictions on the weakly augmented image, $x_{T_m}^{i,w}$, as follows:

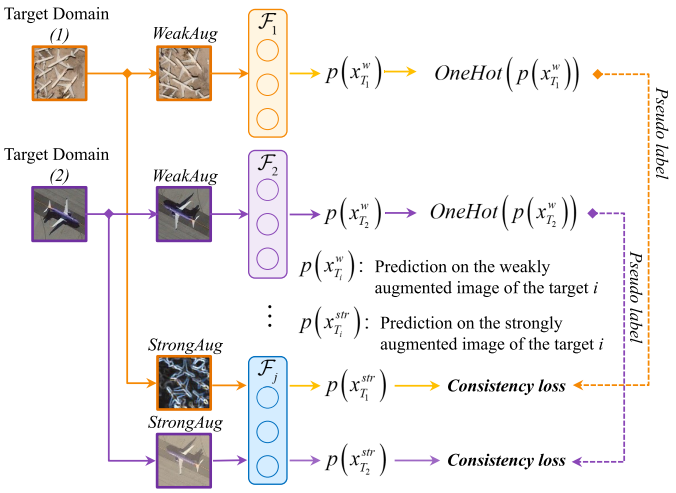


Fig. 4. Illustrating the proposed collaborative strategy's operation.

$$\mathbf{p}_m^{i,w} = p\left(x_{T_m}^{i,w}\right) = \sigma\left(\mathcal{F}_m\left(\mathcal{G}\left(x_{T_m}^{i,w}\right)\right)\right) \tag{13}$$

where $\sigma(\cdot)$ is the *softmax* function; we abbreviate $p(\cdot)$ as $\mathbf{p}$ throughout this article for convenience. These predictions are converted to the one-hot pseudolabel as follows:

$$\hat{y}_m = \text{argmax}\left\{\mathbf{p}_m^{i,w}\right\}, \quad \text{if } \max\left(\mathbf{p}_m^{i,w}\right) > \tau \tag{14}$$

where $\hat{y}_m \in K$ is the pseudolabel created by the mth classifier, $\mathcal{F}_m$. Since the trained model is very sensitive to incorrect pseudolabels, we only select the confident pseudolabel, $\max(\mathbf{p}_m^{i,w}) > \tau$. Second, the jth classifier, $\mathcal{F}_j$, provides its predictions on representations of the strong augmentation image, $x_{T_m}^{i,\text{str}}$, extracted by the feature extractor as follows:

$$\mathbf{q}_j^{i,\text{str}} = p\left(x_{T_m}^{i,\text{str}}\right) = \sigma\left(\mathcal{F}_j\left(\mathcal{G}\left(x_{T_m}^{i,\text{str}}\right)\right)\right) \tag{15}$$

where $\mathbf{q}_j^{i,\text{str}}$ denotes the prediction vector of the strong augmentation image in the mth target domain predicted by the jth classifier. The cost function used to train the classifier $\mathcal{F}_j$ under the supervision of the classifier $\mathcal{F}_m$ is defined as follows:

$$\mathcal{L}_{\mathcal{F}_{j,m}} = \frac{1}{n_{T_m}}\sum_{i=1}^{n_{T_m}} \mathbb{1}\left[\max\left(\mathbf{p}_m^{i,w}\right) > \tau\right] H\left(\hat{y}_m, \mathbf{q}_j^{i,\text{str}}\right) \tag{16}$$

where $H(\cdot)$ is the cross-entropy. To reduce divergence representation distributions between targets m and j, we train the model to minimize $\mathcal{L}_{\mathcal{F}_{j,m}}$.

However, our final goal is to design a joint model in such a way that it can satisfactorily perform knowledge transfer across multiple-target domains. Therefore, we leverage knowledge from multiple domain-specific experts to teach a nonexpert domain via the collaborative strategy. Fig. 4 illustrates the proposed collaborative strategy; this process is

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formulated as follows:

$$\mathcal{L}_{\mathcal{F}_j}^{\text{Col}} = \frac{1}{N-1} \sum_{\substack{m=1,\\ m\neq j}}^{N} \sum_{i=1}^{n_{\mathcal{T}_m}} \mathbb{1}\left[\max\left(\mathbf{p}_m^{i,w}\right) > \tau\right] H\left(\hat{y}_m, \mathbf{q}_j^{i,\text{str}}\right) \quad (17)$$

where $\mathcal{L}_{\mathcal{F}_j}^{\text{Col}}$ is the collaborative loss. Equation (17) explicitly describes how $(N-1)$ domain-specific expert models teach the jth nonexpert domain model. The pseudolabels generated from $(N-1)$ domain-expert classifiers are converted into a one-hot encoded label to minimize the cross-entropy with the prediction of the jth classifier.

The total learning cost functions to train the shared feature extractor are computed as follows:

$$\mathcal{L}_{\mathcal{G}} = \mathcal{L}_{\mathcal{G}}^{\text{Extract}} + \mathcal{L}_{\mathcal{G}}^{\text{MCC}} + \mathcal{L}_{\mathcal{G}}^{\text{Cont}} + \frac{1}{N}\sum_{j=1}^{N} \mathcal{L}_{\mathcal{F}_j}^{\text{Col}}. \quad (18)$$

The cost functions to update the parameters of every classifier are calculated as follows:

$$\mathcal{L}_{\mathcal{F}_j} = \mathcal{L}_{\mathcal{F}_j}^{\text{Extract}} + \mathcal{L}_{\mathcal{F}_j}^{\text{MCC}} + \mathcal{L}_{\mathcal{F}_j}^{\text{Cont}} + \mathcal{L}_{\mathcal{F}_j}^{\text{Col}}. \quad (19)$$

C. Model Inference

We evaluate the proposed method by averaging the *softmax* outputs of different classifiers $\{\mathcal{F}_j\}_{j=1}^{N}$ as follows:

$$\hat{y}_T^i = \text{argmax}\left(\frac{1}{N}\sum_{j=1}^{N} \mathcal{F}_j\left(\mathcal{G}\left(x_T^i\right)\right)\right) \quad (20)$$

where $x_T^i \in \mathcal{D}_T = \{\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_N\}$.

D. Complexity Analysis

We analyze the computational complexity based on components in the proposed method, including the share feature extractor $\mathcal{G}$ and multiple classifiers $\{\mathcal{F}_j\}_{j=1}^{N}$, which is inspired by [31]. Specifically, the computational complexity to find the optimal parameters of the feature extractor during T_1 training iterations is computed as follows:

$$\mathcal{O}\left(T_1 \cdot \sum_{l=1}^{L}\left(M_l^2 \cdot K_l^2 \cdot C_{\text{in}} \cdot C_{\text{out}}\right)\right) \quad (21)$$

where L is the number of convolutional layers of the feature extractor, which has the feature map size M_l^2 and the kernel size K_l^2 with output and input channels C_{int} and C_{out}. Similarly, the computational complexity to optimize the parameters of multiple classifiers during T_2 training iterations is computed as follows:

$$\mathcal{O}\left(T_2 \cdot \sum_{i=1}^{M}\sum_{y=1}^{Y}\left(H_{y-1} \cdot H_y\right)\right) \quad (22)$$

where M and Y are the numbers of target domains and fully connected (FC) layers, respectively, and H_{y-1} is the number of hidden nodes corresponding to the $(y-i)$th FC layer.

From (21) and (22), the training computational complexity of the proposed method is calculated as follows:

$$\mathcal{O}\left(T_1 \cdot \sum_{l=1}^{L}\left(M_l^2 \cdot K_l^2 \cdot C_{\text{in}} \cdot C_{\text{out}}\right)\right) + \mathcal{O}\left(T_2 \cdot \sum_{i=1}^{M}\sum_{y=1}^{Y}\left(H_{y-1} \cdot H_y\right)\right). \quad (23)$$

IV. Experiment

In this section, we first describe the five RS scene datasets widely used in detail. Second, we provide comprehensive experiments compared to SOTA methods on these datasets to evaluate our method. Finally, we conduct ablation studies in Section IV-E to clarify the effectiveness of each module in the EHACA approach.

A. Datasets' Description

The proposed method's effectiveness for MTDA is evaluated by conducting it on five RS datasets: **AID** (**A**) [32], **NWPU-RESISC45** (**N**) [33], **PatternNet** (**P**) [34], **RSD46-WHU** (**R**) [35], and **UC Merced** (**U**) [36].

1) The **AID** (**A**) dataset consists of 10 000 RGB images collected from *Google Earth* imagery by aircraft and satellites, such as *WorldView-2*, *GeoEye-1*, and *Landsat 8*. These images are categorized into 30 aerial scene classes.
2) The **NWPU-RESISC45** (**N**) dataset contains 31 500 images extracted from *Google Earth* imagery that are classified into 45 scene classes.
3) The **PatternNet** (**P**) dataset is collected from high-resolution *Google Earth* imagery. This dataset covers 38 scene classes with 800 RGB images.
4) The **RSD46-WHU** (**R**) dataset is a large-scale dataset containing a total of 117 000 RGB images collected by hand from *Tianditu* and *Google Earth* and categorized into 46 different classes. Each class contains 500–3000 images.
5) The **UC Merced** (**U**) dataset includes 2100 RGB images, which capture multiple U.S. regions from the United States Geological Survey (USGS) National Map. These images are classified into 21 classes, and each class obtains 100 images.

We tested our method on the collected datasets as in [18][1] for a fair comparison. These datasets share ten classes, namely, *agriculture*, *airfield*, *forest*, *overpass*, *parking*, *port*, *residential*, *sports court*, *storage tank*, and *water*, where several classes are merged from multiple classes that share common features. For example, images in *airport* and *airplane* classes are unified into the same class as *airfield*. Table I and Fig. 5 describe these datasets in detail.

[1] https://github.com/rs-dl/TSAN

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TABLE I
DETAILED INFORMATION OF FIVE RS SCENE DATASETS

Dataset	AID	NWPU-RESISC45	PatternNet	RSD46-WHU	UC Merced
Total images	10,000	31,500	30,400	117,000	2,100
Number of classes	31	45	38	46	21
Images per classes	220-420	700	800	500-3,000	100
Resolution (m)	0.5-8	0.2-30	0.062-4.693	0.5-2	0.3
Size	600 × 600	256 × 256	256 × 256	256 × 256	256 × 256
Source	Google Earth	Google Earth	Google Earth, Google Map	Google Earth, Tianditu	USGS

B. Parameter Setting

We implemented EHACA using the PyTorch framework [40] on a single GeForce RTX3090 GPU with 24-GB RAM. The stochastic gradient descent (SGD) was used as the optimizer to find the optimal parameters of the shared feature extractor and multiple classifiers. The momentum was set as 0.9, and the strategy to adjust the learning rate η was conducted as in [14], $\eta = (\eta_0/((1 + 10p)^{0.75}))$, where $p \in [0, 1]$ and the initial learning rate $\eta_0 = 0.001$. The weight of decay was set to 0.0005. For a fair comparison with the SOTA approaches, we used AlexNet [41] and ResNet-50 [42] as the feature extractor pretrained on the ImageNet dataset [46]. The classifier associated with AlexNet consists of a single FC layer, while the classifier corresponding to ResNet-50 contains two linear FC layers. The batch size was set to 96 for the AlexNet backbone but 48 for the ResNet-50 backbone considering GPU memory constraint.

C. Comparison With the State-of-the-Art Methods

To verify the effectiveness of EHACA, we compared it with previous SOTA methods. These methods are divided into two groups called S^{3}TDA and multiple-target domains adaptation with the following purposes. First, we demonstrated that the S^{3}TDA methods were challenging to extend to the MTDA setting. Second, we showed that the proposed method addressed the existing limitations of the previous multiple-target domain approaches by providing outstanding classification performance.

The S^{3}TDA methods included the following:

1) deep adaptation network (DAN) [37];
2) domain-adversarial neural network (DANN) [14];
3) joint adaptation networks (JANs) [38];
4) conditional domain adversarial networks (CDANs) [39].

The MTDA methods included the following:

1) a two-stage adaptation network (TSAN) [18];
2) D-CGCT [22].

We extracted all results of the SOTA approaches using the released codes[2,3] in the same environment as the proposed method for a fair comparison, except for the results of TSAN, which were collected from previous work [18].

D. Result Analysis

We analyzed the classification results of the proposed EHACA and benchmark methods over five DA tasks. These classification results were divided into *single-target* (*S–T*), *target-combined* (*T–C*), and *multitarget* (*M–T*) settings. The classification results over multiple-target domains under the *S-T* setting were calculated as follows:

$$\mathrm{ACC}_{S-T_i} = \mathrm{Model}_i(T_i) \tag{24}$$

where Model_i is used to extract the classification result of the target T_i in the S^{3}TDA task, $S \rightarrow T_i$. We need to train each model corresponding to each domain task for adaptation. For example, the classification accuracy of the **N**→{**A**, **P**, **R**, **U**} task was calculated by taking the average classification results of four S^{3}TDA tasks, such as **N**→**A**, **N**→**P**, **N**→**R**, and **N**→**U**. In contrast, the classification accuracy of multiple-target domains in the *T–C* setting was calculated by performing a single-source to mixed target domains as follows:

$$\mathrm{ACC}_{T-C} = \mathrm{Model}(T_{\mathrm{mixed}}). \tag{25}$$

The classification accuracy of the *M–T* setting was calculated using a one-joint model for adaptation to multiple-target domains as follows:

$$\mathrm{ACC}_{\mathrm{MT}_i} = \mathrm{Model}(T_i). \tag{26}$$

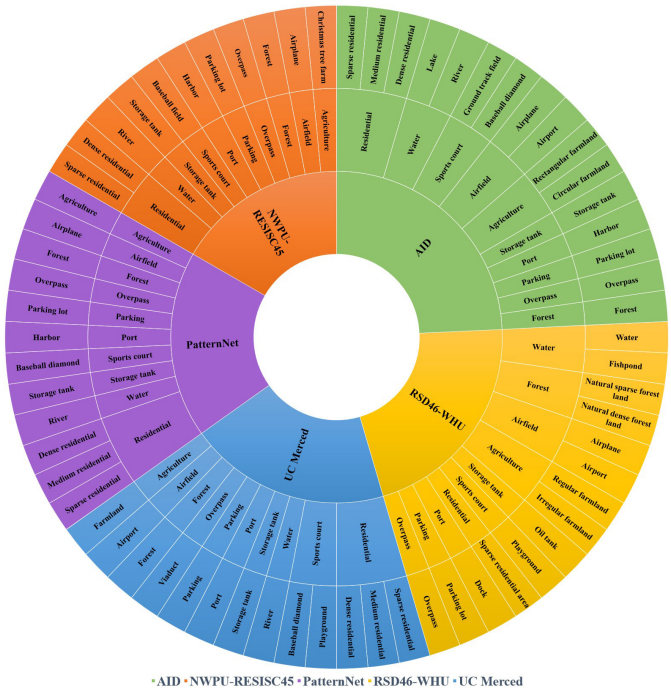


Fig. 5. Hierarchical graph of shared ten classes over five RS datasets.

[2]https://github.com/thuml/Transfer-Learning-Library
[3]https://github.com/Evgeneus/Graph-Domain-Adaptaion

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TABLE II

ACCURACY (%) OF THE PROPOSED AND BENCHMARK METHODS OVER FIVE DA TASKS BASED ON THE ALEXNET NETWORK

Setting	Source / Target	A				N				P				R				U				Avg.
		N	P	R	U	A	P	R	U	A	N	R	U	A	N	P	U	A	N	P	R	(%)
S–T	DAN	80.8	80.5	67.2	72.3	86.8	87.9	58.4	82.3	66.2	68.5	50.5	82.7	69.0	60.9	67.6	62.0	67.0	61.8	91.2	50.0	70.7
	DANN	83.8	84.3	70.8	82.6	86.7	89.4	62.2	84.8	68.2	**72.8**	**55.0**	92.9	74.1	65.1	**72.6**	**69.3**	67.7	65.6	**92.4**	47.7	74.4
	JAN	82.5	**90.8**	70.5	75.2	86.0	**91.3**	56.2	83.6	71.6	70.1	54.7	89.3	70.9	64.9	64.7	67.4	**72.3**	69.2	91.3	**52.9**	73.8
	CDAN	**86.3**	88.4	**77.9**	**86.3**	**91.0**	89.2	**69.6**	**90.6**	**73.5**	71.6	50.2	**95.0**	**75.9**	**68.5**	71.9	68.8	68.6	**71.0**	91.2	50.4	**76.8**
T–C	DAN	73.1				69.3				54.3				62.9				58.2				63.6
	DANN	75.0				71.3				57.2				66.5				61.4				66.3
	JAN	74.8				69.4				**62.8**				64.2				**67.6**				67.8
	CDAN	**76.1**				**75.7**				59.7				**67.6**				63.5				**68.5**
	TSAN	70.6				69.2				53.9				64.4				58.2				63.3
M–T	D-CGCT	90.4	90.8	76.7	**96.6**	**93.1**	87.4	71.6	86.8	64.8	56.5	42.7	75.4	78.7	78.8	**92.0**	**81.3**	73.8	68.1	92.3	50.6	77.4
	EHACA	**95.4**	**91.4**	**93.0**	94.5	92.7	**99.8**	**92.0**	**96.1**	**87.1**	**84.2**	**88.5**	**97.9**	**83.0**	**79.8**	84.8	80.0	**85.3**	**84.3**	**99.8**	**83.3**	**89.6**

Note: S–T: Single-Target; T–C: Target-Combined; M–T: Multi-Target

TABLE III

ACCURACY (%) OF THE PROPOSED AND BENCHMARK METHODS OVER FIVE DA TASKS BASED ON THE RESNET-50 NETWORK

Setting	Source / Target	A				N				P				R				U				Avg.
		N	P	R	U	A	P	R	U	A	N	R	U	A	N	P	U	A	N	P	R	(%)
S–T	DAN	84.2	82.6	68.5	77.6	89.0	92.2	65.9	82.9	66.5	69.3	52.9	83.4	69.2	64.9	67.7	63.8	68.3	62.5	94.0	53.4	72.9
	DANN	89.4	90.1	77.9	92.1	92.7	91.0	69.2	88.4	**77.9**	75.6	55.7	94.5	80.9	78.9	80.5	72.9	81.0	73.7	**93.9**	53.9	80.5
	JAN	88.1	91.0	72.2	77.8	89.4	92.0	66.8	86.0	77.0	**79.8**	**59.3**	90.9	73.3	76.0	75.9	70.3	75.6	**78.5**	92.5	**62.3**	78.7
	CDAN	**95.0**	**97.4**	**81.2**	**92.3**	**97.1**	**93.3**	**84.1**	**96.2**	**77.9**	78.5	58.3	**98.1**	**83.4**	**79.7**	**81.2**	**73.3**	**85.5**	75.1	93.1	57.0	**83.9**
T–C	DAN	76.3				74.0				55.1				68.0				60.6				66.8
	DANN	79.7				76.5				61.7				74.8				63.3				71.2
	JAN	78.5				73.0				**75.3**				69.4				71.6				73.5
	CDAN	84.1				79.1				67.9				**76.6**				**71.7**				**75.9**
	TSAN	**84.3**				**81.9**				60.5				73.4				68.1				73.6
M–T	D-CGCT	96.6	99.3	84.2	**98.9**	**98.1**	97.5	81.0	**98.9**	80.2	76.3	69.5	91.8	91.2	84.1	92.4	85.9	88.5	77.7	96.6	75.4	88.2
	EHACA	**97.7**	**99.9**	**97.0**	97.6	97.2	**99.9**	**97.1**	98.8	**89.9**	**90.2**	**93.0**	**98.4**	**97.0**	**96.0**	**99.6**	**97.3**	**97.2**	**90.4**	**99.8**	**91.5**	**96.3**

Tables II and III report the comparison classification results of different benchmark methods and the proposed approach over five DA tasks under the *S–T*, *T–C*, and *M–T* settings. As shown in these tables, the average classification accuracy of the methods designed for the *M–T* setting achieved outstanding performance compared to results extracted by both *S–T* and *T–C* settings.

The average classification accuracy of the proposed method recorded the highest result, namely, 89.6% and 96.3% for AlexNet and ResNet-50 backbones, respectively. The average classification accuracy of EHACA was improved by 12.2% and 8.1% compared to the second-best approach, D-CGCT, for AlexNet and ResNet-50, respectively. CDAN [39] reported the best classification performance in both *S–T* and *T–C* settings. However, the average classification results of CDAN under *S–T* and *T–C* settings were still lower than 12.8% and 21.1% for the averaging classification results of the proposed method, respectively, extracted by the AlexNet backbone network. Similarly, for the ResNet-50 backbone network, the classification results of the proposed EHACA were higher than 12.4% and 20.4% compared to the averaged classification results of the CDAN method under *S–T* and *T–C* settings, respectively.

The classification results tabulated in Tables II and III revealed two important sides in the multiple-target domains. First, the existing S^{3}TDA approaches were difficult to extend to the multiple-target domains adaptation setting. Second, as mentioned in Section I, the semantic information of various target domains can be damaged when we try to merge them into a single-target domain; therefore, it negatively affected the discriminative ability of the trained model. Consequently, the average classification results under the *T–C* setting were lower than that of the *S–T* setting.

E. Ablation Study

As discussed in Section III, the proposed EHACA includes *Extracting source representations*, *minimizing class confusion*, *Contrasting*, and *Collaborating*. In this section, we provided the experimental results for evaluating the influence of each component on the proposed method. Besides, we provided an analysis of the negative transfer effect in the multitarget domain setting.

1) Impact of Each Component: Table IV lists the classification results for analyzing the impact of each component in the proposed method over five DA tasks based on the AlexNet backbone network. In the extraction step, we trained the *baseline* that consists of the shared feature extractor and multiple classifiers ($N = 4$) by only using the labeled source samples. Equations (1) and (2) describe the cost function in this step ($\mathcal{L}^{\text{Extract}}$), where the loss functions (1) was used to update the parameters of each classifier, while the loss function (2) was used to find the optimal parameters of the

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TABLE IV

CLASSIFICATION ACCURACY (%) OF THE PROPOSED METHOD FOR ABLATION STUDY TO EVALUATE THE IMPACT OF EACH COMPONENT IN THE MULTIPLE-TARGET DOMAIN SETTING BASED ON THE ALEXNET BACKBONE NETWORK

Loss Function				Domain Adaptation Task					Avg. (%)
$\mathcal{L}^{Extract}$	$\mathcal{L}^{MCC}$	$\mathcal{L}^{Cont}$	$\mathcal{L}^{Col}$	**A**→{**N**,**P**,**R**,**U**}	**N**→{**A**,**P**,**R**,**U**}	**P**→{**A**,**N**,**R**,**U**}	**R**→{**A**,**N**,**P**,**U**}	**U**→{**A**,**N**,**P**,**R**}	
✓				72.09	68.68	49.52	62.40	59.71	62.48
✓	✓			81.36	73.53	53.41	74.79	64.63	69.54
✓	✓	✓		85.40	85.37	69.85	80.17	72.61	78.68
✓	✓	✓	✓	93.58	95.14	89.44	81.86	88.16	89.64

TABLE V

CLASSIFICATION ACCURACY (%) UNDER DIFFERENT DA SETTINGS FOR ABLATION STUDY OF NEGATIVE TRANSFER

Network	Setting	Domain Adaptation Task					Avg. (%)
		A→{**N**,**P**,**R**,**U**}	**N**→{**A**,**P**,**R**,**U**}	**P**→{**A**,**N**,**R**,**U**}	**R**→{**A**,**N**,**P**,**U**}	**U**→{**A**,**N**,**P**,**R**}	
AlexNet	Baseline	72.09	68.68	49.52	62.40	59.71	62.48
	T–C	82.46	77.26	50.88	63.94	63.07	67.52
	M–T (EHACA)	93.58	95.14	89.44	81.86	88.16	89.64
ResNet-50	Baseline	79.98	75.02	51.79	66.39	64.40	67.52
	T–C	88.36	89.51	63.33	86.82	80.39	81.68
	M–T (EHACA)	98.03	98.24	92.86	97.47	94.75	96.27

TABLE VI

ABLATION STUDY TO EVALUATE THE NEGATIVE EFFECT OF THE PROPOSED METHOD VIA TWO TYPE INDEXES INCLUDING THE ABSOLUTE NEGATIVE TRANSFER EFFECT (ANTE%) AND THE RELATIVE NEGATIVE TRANSFER EFFECT (RNTE%)

Network	Domain Adaptation Task										Avg. (%)	
	A→{**N**,**P**,**R**,**U**}		**N**→{**A**,**P**,**R**,**U**}		**P**→{**A**,**N**,**R**,**U**}		**R**→{**A**,**N**,**P**,**U**}		**U**→{**A**,**N**,**P**,**R**}			
	ANTE	RNTE	ANTE	RNTE	ANTE	RNTE	ANTE	RNTE	ANTE	RNTE	ANTE	RNTE
AlexNet	10.37	-10.75	8.58	-17.88	1.36	-38.56	1.54	-17.92	3.36	-25.09	5.04	-22.04
ResNet-50	8.38	-9.67	14.49	-8.73	11.54	-29.53	20.43	-10.65	15.99	-14.36	14.17	-14.59

shared feature extractor. The average classification in the extraction step only achieved 62.48%. However, when we added the minimizing class confusion loss in (9) and (10) ($\mathcal{L}^{\text{Extract}} + \mathcal{L}^{\text{MCC}}$), it reduced class confusion in the target domain, so the average classification was improved by 7.06%. Then, the trained model became more robust by conducting contrastive loss in (11) and (12) ($\mathcal{L}^{\text{Extract}} + \mathcal{L}^{\text{MCC}} + \mathcal{L}^{\text{Cont}}$); thus, the average classification loss significantly increased to reach 78.68%. Finally, the overall loss function with the collaborating loss ($\mathcal{L}^{\text{Extract}} + \mathcal{L}^{\text{MCC}} + \mathcal{L}^{\text{Cont}} + \mathcal{L}^{\text{Col}}$) boosted the average classification results over five DA tasks to 89.64%. The collaborative strategy allowed different domain-expert classifiers to jointly explore common semantic information across all target domains to provide a well-generalized model.

2) Negative Transfer Effect: In the MTDA setting, domain shift exists not only in each source and target pair but also among different target domains. Therefore, the negative transfer phenomenon occurs when we use the model trained on the source domain to adapt to the target domain directly or combine all target domains as a single-target domain for training. To evaluate the negative transfer effect under different settings: *baseline*, *T–C*, and *M–T* (EHACA), we used two indexes such as the ANTE and the RNTE as in [18].

The classification accuracy on the target domains of the *baseline* setting was extracted by using the model trained on the source domain without using any DA strategy. We modified the proposed method to make it suitable for collecting results of the *T–C* setting. The feature extractor $\mathcal{G}$ and a single classifier $\mathcal{F}_1$ were trained on the labeled source samples as in (1), while the information of combined unlabeled samples was extracted as in (9), (11), and (16). However, only a single classifier was available in the *T–C* setting; thus, (16) was rewritten as follows:

$$\mathcal{L}_{\mathcal{F}_1}^{\text{self}} = \frac{1}{n_{T_{\text{mixed}}}} \sum_{i=1}^{n_{T_{\text{mixed}}}} \mathbb{1}\left[\max\left(\mathbf{p}_1^{i,w}\right) > \tau\right] H\left(\hat{y}_1, \mathbf{q}_1^{i,\text{str}}\right) \quad (27)$$

where $\mathcal{L}_{\mathcal{F}_1}^{\text{self}}$ is the cost function of a self-training strategy. $\hat{y}_1 = \text{argmax}\{\mathbf{p}_1^{i,w}\}$, if $\max(\mathbf{p}_1^{i,w}) > \tau$, and $n_{T_{\text{mixed}}}$ was the number of samples of mixed unlabeled target domains. The classification results under the *M–T* setting were extracted using the proposed EHACA.

The ANTE index indicated the negative transfer level when directly applying the trained model on the source domain to the target domains by comparing the averaging classification between the *T–C* and the *baseline* settings, which was calculated as follows:

$$\text{ANTE} = \text{ACC}_{T-C} - \text{ACC}_{\text{Baseline}} \quad (28)$$

where ACC_{T-C} was the classification accuracy of the *T–C* setting and $\text{ACC}_{\text{Baseline}}$ was the classification accuracy of the

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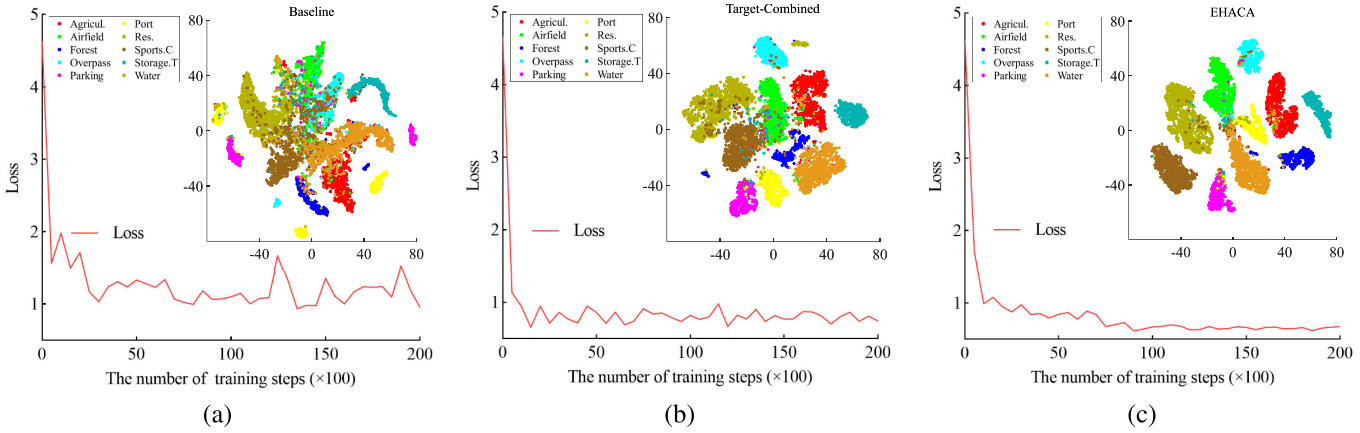


Fig. 6. Ablation study of the negative transfer phenomenon. We extracted presentations and testing losses from the target domain **NWPU-RESISC45** in the DA task **A**→{**N**, **P**, **R**, **U**} under different settings using AlexNet as the backbone network. (a) Extracted results from the *baseline*. (b) Reported results under the *target-combined* setting. (c) Results of the proposed EHACA under the *multitarget* setting.

baseline setting. Besides, the RNTE was used to estimate the negative transfer that occurred when combining all target domains, which was computed by comparing the classification accuracy between the *T–C* and *M–T* settings as follows:

$$\text{RNTE} = \text{ACC}_{T-C} - \text{ACC}_{M-T} \tag{29}$$

where ACC_{M-T} was the classification accuracy of the *M–T* setting. Table V showed the classification accuracy of the different settings consisting of *baseline*, *T–C*, and *M–T* (EHACA), respectively. Then, we reported the negative transfer effect results calculated by following (28) and (29) in Table VI. As shown in this table, the ANTE indexes received a positive value for all DA tasks, which indicated that the negative transfer effect was serious if we directly employed the *baseline* model on the target domain. Specifically, the average classification result of the *T–C* setting was improved by 5.04% for AlexNet and 14.17% for ResNet-50 compared to the *baseline* setting; these classification results were concordant with example visualization results illustrated in Fig. 6(a) and (b). As shown in these figures, the testing loss of the target domain **NWPU-RESISC45** (**N**) on the DA task **A**→{**N**, **P**, **R**, **U**} under the *T–C* setting was lower than that of the *baseline* setting. Consequently, the representations of this target domain under the *T–C* setting were more discriminative than the *baseline* setting.

In contrast to the ANTE index, the RNTE indexes received a negative value over five DA tasks, revealing that the learned model struggled to classify the representations of the mixed target data because of increasing data variance. Besides, the proposed EHACA under the *M–T* setting was less affected by the negative transfer because we used the easy-to-hard structure with the divide-to-adapt strategy that could reduce the data variance and explicitly extract the private information in each target domain. In addition, in the *M–T* setting, we proposed a collaborative strategy that encouraged different expert models to exchange their knowledge to exploit common semantic information from different target domains effectively; thus, the average classification accuracy of the proposed EHACA was 22.11% and 14.59% higher than the *T–C* setting for the AlexNet and ResNet-50 backbone networks, respectively. As shown in Fig. 6, the target domain **NWPU-RESISC45** (**N**) testing loss on the DA task **A**→{**N**, **P**, **R**, **U**} under the *M–T* setting was the lowest; therefore, the target representations extracted by the proposed method were the most distinguishable, as shown in Fig. 6(c).

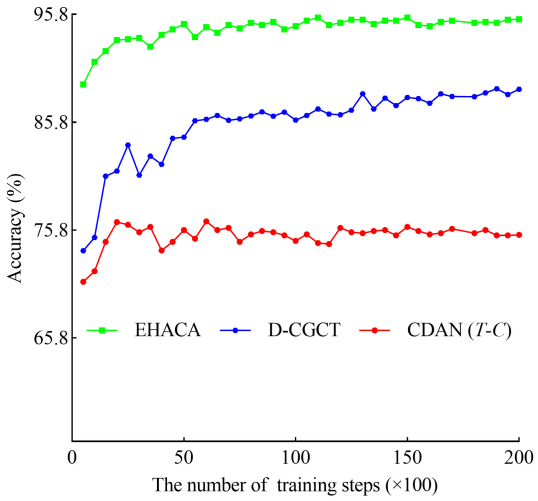


Fig. 7. Convergence analysis on **A**→{**N**, **P**, **R**, **U**} using the AlexNet as the backbone network.

F. Convergence Analysis

Fig. 7 illustrated the classification accuracy along with the training steps of different DA methods, including CDAN (*T–C*), D-CGCT, and EHACA on the DA task **A**→{**N**, **P**, **R**, **U**} using the AlexNet as the backbone network. It is obvious that the classification accuracy of the proposed EHACA quickly converged after 60 000 training steps, and it maintained stably by 95.3% until the end of the training step. In contrast, CDAN (*T–C*) showed the lowest performance in the target classification accuracy, which fluctuated during 150 000 training steps with the highest accuracy of 76.1%. Then, it remained stable at around 75.3%. This was because,

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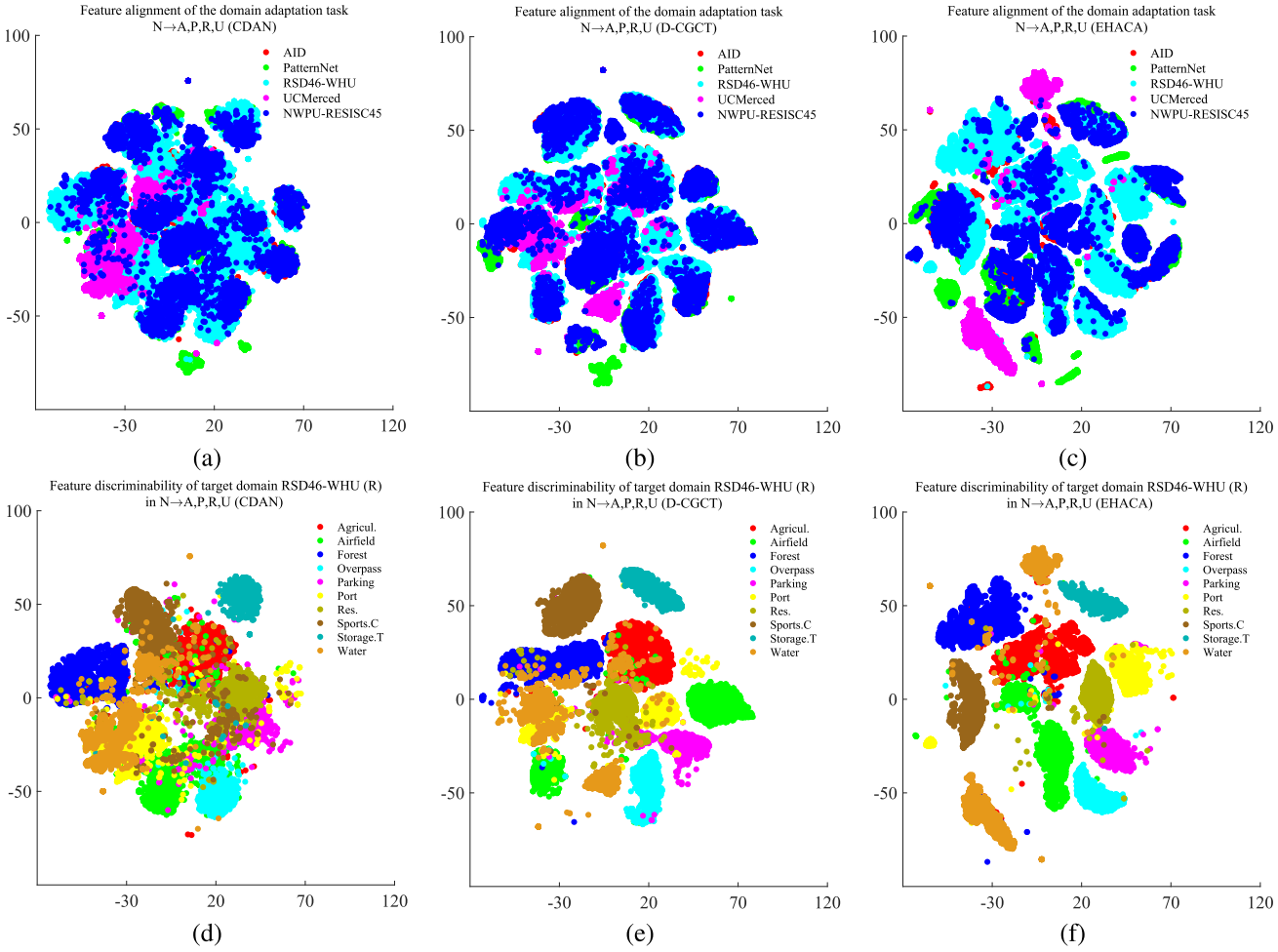


Fig. 8. Feature alignment and feature discriminability visualizations of the different methods on the DA task **N**→{**A**, **P**, **R**, **U**} by using t-SNE [43], based on ResNet-50. (a)–(c) Representations of CDAN, D-CGCT, and EHACA in terms of domain feature alignment (*domain level*). (d)–(f) Examples of representations extracted by CDAN, D-CGCT, and EHACA in terms of feature discriminability (*class level*) on the target domain **RSD46-WHU** (**R**).

in the CDAN (*T*–*C*) setting, all target domains were combined into a single target; therefore, the learned model struggled to classify the semantic information from different domains due to the increased data variance. Moreover, CDAN only focused on domain alignment without considering distinguishable representations in each class within the target domain and among various target domains.

The classification performance of D-CGCT kept growing through the training steps, demonstrating the effectiveness of the easy-to-hard structure and the divide-to-adaptation training strategy. However, unlikely the proposed EHACA, D-CGCT only emphasized the representation alignment between a source–target pair without considering the correlation among different target domains. Therefore, this method struggled to converge compared to EHACA.

G. Visualization Analysis

1) Feature Visualization: The knowledge transferability from the source domain to multiple-target domains could be evaluated via two aspects called feature alignment (*domain level*) and feature discriminability (*class level*). CDAN showed the highest classification results under *S*–*T* and *T*–*C* settings, as shown in Tables II and III, while the recent MTDA approach, D-CGCT, achieved outstanding results under the *M*–*T* setting. Therefore, we selected the CDAN and D-CGCT methods for comparison to the proposed method. Fig. 8(a)–(c) illustrated the feature alignment of the different methods, CDAN, D-CGCT, and EHACA on the DA task **N**→{**A**, **P**, **R**, **U**} based on ResNet-50 using t-SNE [43]. As shown in these figures, the CDAN and D-CGCT methods provided good domain alignment results compared to the proposed method. However, the CDAN and D-CGCT methods struggled to distinguish between target representations compared to the proposed EHACA, as shown in Fig. 8(d)–(f) on the example target domain **RSD46-WHU** (**R**). This is because the CDAN method relied heavily on reducing the domain discrepancy but did not consider class-level discriminability. Indeed, a recent method [15] demonstrated that, even if the domain alignment level was improved, the class-level discriminability worsened due to negative transfer. Thus, to achieve both feature alignment and class-level discriminability, D-CGCT utilized CDAN to minimize domain discrepancy in each source and target pair, while it used the graph convolutional network (GCN) to aggregate samples, having similar representations for class-level discriminability across domains. However, the D-CGCT only

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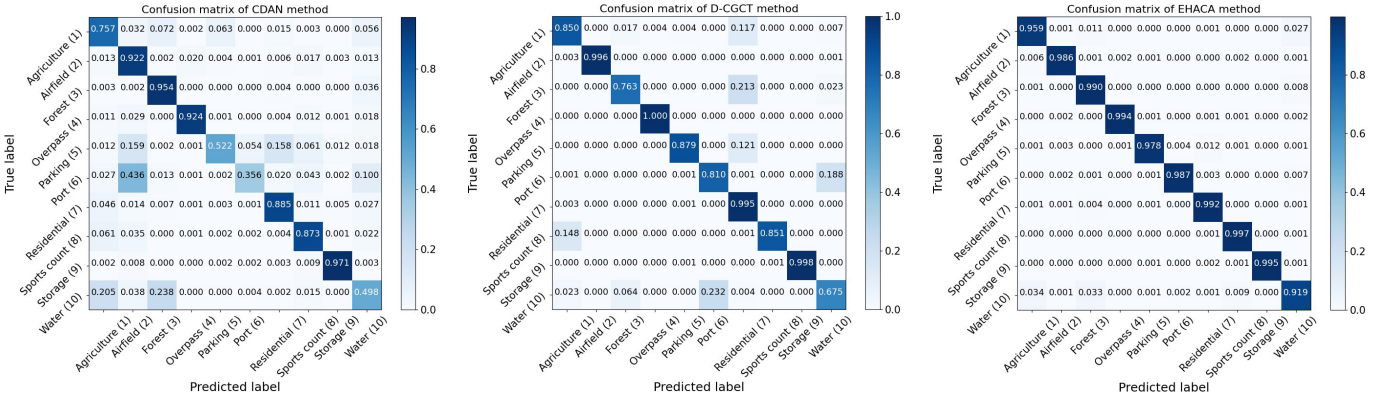


Fig. 9. Visualization results for the confusion matrix; these results were extracted from the **N**→{**A**, **P**, **R**, **U**} DA task based on the ResNet-50 network. (a) CDAN. (b) D-CGCT. (c) EHACA.

exploited the correlation of source–target pairs and ignored exploring the relationship among different target domains. In contrast, the proposed method not only considered feature alignment and feature discriminability in each source and target pair but also explored different target domains; thus, the trained model could jointly perform satisfactorily across all target domains. We could observe that the proposed method's ability to extract representations of the target domain **RSD46-WHU** (**R**) was the most distinguishable, as shown in Fig. 8(f).

2) Confusion Matrix Visualization: Fig. 9 showed the confusion matrix for predictions of CDAN, D-CGCT, and EHACA on the DA task **N**→{**A**, **P**, **R**, **U**} based on ResNet-50. The CDAN method showed the most class confusion due to negative transfer, as shown in Fig. 9(a). The D-CGCT method reduced the data variance by training each source–target pair separately. However, the trained model struggled to achieve generalization on all target domains because the correlation among multiple-target domains was not considered. Therefore, this method still encountered difficulties when distinguishing class pairs, such as {*Port* and *Water*} and {*Forest* and *Residential*}, as shown in Fig. 9(b). The proposed EHACA showed outstanding performance compared to both CDAN and D-CGCT methods. However, several confusions existed among the different classes of *Agriculture*, *Forest*, and *Water* because these classes shared semantic features, as shown in Fig. 9(c).

3) Attention Map Visualization: Fig. 10 visualized the attention maps of CDAN, D-CGCT, and the proposed EHACA. The attention map was extracted from the last convolutional layer of ResNet-50 on the DA task **N**→{**A**, **P**, **R**, **U**} by using Grad-CAM [45]. The red color indicates the region of interest using the trained model. We randomly selected the input images in three classes, *Port*, *Forest*, and *Parking*, from four target domains **A**, **P**, **R**, and **U**. As shown in Fig. 10, our method could extract semantic representations more precisely compared to the CDAN and D-CGCT methods. Specifically, as shown in the *Port* and *Forest* classes, the proposed method focused on the accurate region. In contrast, both CDAN and D-CGCT struggled to distinguish between the class pairs {*Port* and *Water*} and {*Forest* and *Residential*}. In addition,

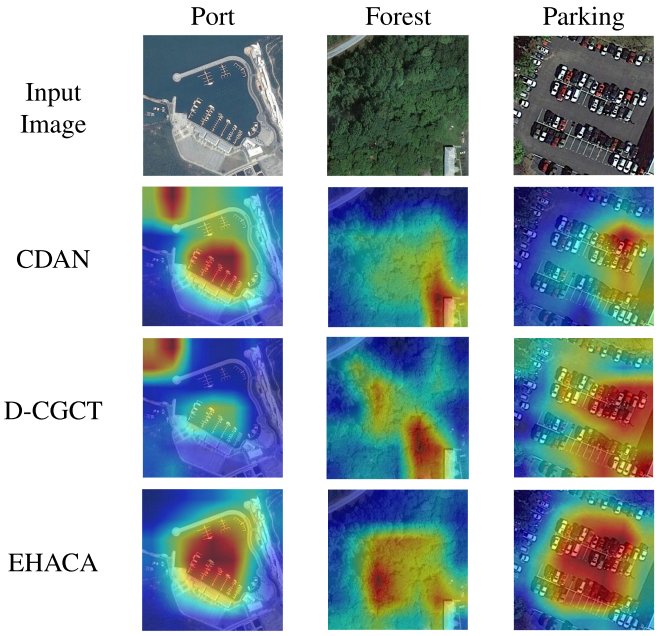


Fig. 10. Attention map visualization results using Grad-Cam [45]. These visualization results were extracted from the final convolution layer of ResNet-50 on the DA task **N**→{**A**, **P**, **R**, **U**}. The red color indicates the region of interest marked by the trained model. The regions of less interest to the trained model are colored blue.

we observed that the proposed method was more concentrated on the critical region in image classification compared to the CDAN and D-CGCT methods as in the extracted visualization result of the *Parking* class. These extracted results demonstrated that the proposed method was well-adapted to the *multitarget* DA task.

V. Conclusion

This article introduced a novel MTDA method for RS sense classification based on the proposed easy-to-hard adaption structure called EHACA. This structure equips three components: extracting source representations, hierarchical intratarget feature alignment, and collaborative intertarget feature

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alignment, used to extract the training data information from easy to hard fashion, respectively, for enhancing classification performance. First, the model easily extracted information from the source data using label information in the extracting step. Second, the trained model tried to encode the unique characteristic in the particular target domain using the hierarchical intratarget feature alignment strategy. Finally, the trained model could jointly perform all target domains in the collaborating strategy. The experimental results over five different RS sense classification datasets demonstrated the effectiveness of our method. The classification accuracy of the proposed method surpassed the current MTDA benchmark methods to achieve a SOTA MTDA approach.

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